

2.2.4 PHYSICAL SCIENCE STREAM

2.2.4.1 Engineering (Course Code - 008) (Proposed Intake - 2238)

Minimum eligibility requirements for admission :

At least three 'S' grades in Chemistry, Combined Mathematics and Physics at the G.C.E. (Advanced Level) Examination.

Fields of specialization available in each university are :

University of Peradeniya:

- Civil Engineering
- Chemical & Process Engineering
- Computer Engineering
- Electrical & Electronic Engineering
- Mechanical Engineering
- Manufacturing & Industrial Engineering

University of Moratuwa:

- Biomedical Engineering
- Electrical Engineering
- Chemical & Process Engineering
- Electronic and Telecommunication Engineering
- Civil Engineering
- Mechanical Engineering
- Computer Science & Engineering
- Materials Science & Engineering

University of Ruhuna:

- Civil and Environmental Engineering
- Electrical and Information Engineering
- Mechanical and Manufacturing Engineering
- Marine Engineering & Naval Architecture
- Computer Engineering

University of Jaffna:

- Civil Engineering
- Electrical and Electronic Engineering
- Mechanical Engineering
- Computer Engineering

South Eastern University of Sri Lanka:

- Civil Engineering
- Electrical Engineering #
- Electronic Engineering #
- Mechanical Engineering

For the time being Electrical Engineering and Electronic Engineering will be combined and offered as 'Electrical & Electronic Engineering'.

University of Sri Jayewardenepura:

- Civil Engineering
- Computer Engineering
- Electrical and Electronic Engineering
- Mechanical Engineering

Selection for a particular field of specialization specified above would be done by the University to which the candidate has been admitted.

**2.2.4.2 Engineering (EM) - Earth Resources Engineering
(Course Code - 009)
(Proposed Intake - 80)**

Minimum eligibility requirements for admission :

At least three 'S' grades in Chemistry, Combined Mathematics and Physics at the G.C.E. (Advanced Level) Examination.

- Degree Programme : Bachelor of the Science of Engineering Honours in Earth Resources Engineering
- Available University : University of Moratuwa
- Duration : 04 years

**2.2.4.3 Engineering (TM) - Textile & Apparel Engineering
(Course Code - 010)
(Proposed Intake - 98)**

Minimum eligibility requirements for admission :

At least three 'S' grades in Chemistry, Combined Mathematics and Physics at the G.C.E. (Advanced Level) Examination.

- Degree Programme : Bachelor of the Science of Engineering Honours in Textile & Apparel Engineering
[BScEngHons (Textile & Apparel Engineering)]
- Available University : University of Moratuwa
- Duration : 04 years

**2.2.4.4 Transport Management & Logistics Engineering (TMLE)
(Course Code - 057)
(Proposed Intake - 91)**

Minimum eligibility requirements for admission :

At least three 'S' grades in Chemistry, Combined Mathematics and Physics at the G.C.E. (Advanced Level) Examination.

- Degree Programme : Bachelor of the Science of Engineering Honours in Transport Management & Logistics Engineering
[BScEngHons (Transport Management & Logistics Engineering)]

- Available University : University of Moratuwa
- Duration : 04 years

2.2.4.5 Physical Science
(Course Code - 013)
(Proposed Intake - 2262)

Minimum eligibility requirements for admission :

At least 'S' grades in following three subjects at the G.C.E. (Advanced Level) Examination and the three subjects must include;

1. Combined Mathematics or Higher Mathematics
and
2. Chemistry or Physics
and
3. The third subject from the list given below;
 - Agricultural Science
 - Combined Mathematics
 - Biology
 - Higher Mathematics
 - Chemistry
 - Physics

- Degree Programme : Bachelor of Science in Physical Science
- Available Universities : University of Colombo
University of Peradeniya
University of Sri Jayewardenepura
University of Kelaniya
University of Jaffna
University of Ruhuna
Eastern University, Sri Lanka
South Eastern University of Sri Lanka

2.2.4.6 Computer Science
(Course Code - 012)
(Proposed Intake - 686)

Minimum eligibility requirements for admission :

At least a 'C' grade in Combined Mathematics or Physics or Higher Mathematics;
and

At least 'S' grades in two other subjects listed below at the G.C.E. (Advanced Level) Examination;

- Combined Mathematics
 - Higher Mathematics
 - Mathematics
 - Physics
 - Chemistry
 - Information & Communication Technology
- Degree Programme : Bachelor of Science in Computer Science

- Available Institutions : University of Colombo School of Computing (UCSC)
University of Jaffna
University of Ruhuna
Trincomalee Campus, Eastern University, Sri Lanka
- Duration : 03 years
- Medium : English

In addition, following Universities/ Institutes offer 04 year Honours degrees:

- Bachelor of Science Honours in Computer Science
Available Institute : University of Colombo School of Computing (UCSC)
- Bachelor of Science Honours in Software Engineering
Available Institute : University of Colombo School of Computing (UCSC)
- Bachelor of Science Honours in Computer Science [BScHons (ComputerSc)]
Available University : University of Kelaniya
- Bachelor of Science Honours in Computer Science [BScHons (Computer Science)]
Available University : University of Jaffna
- Bachelor of Computing Honours in Computer Science [BCompHons (Comp Sc)]
Available University : University of Sri Jayewardenepura

2.2.4.7 Applied Sciences (Physical Science) **(Course Code - 015)** **(Proposed Intake - 778)**

Minimum eligibility requirements for admission :

At least 'S' grades in following three subjects at the G.C.E. (Advanced Level) Examination and the three subjects must include;

1. Combined Mathematics or Higher Mathematics
and
2. Chemistry or Physics
and
3. The third subject from the list given below;
 - Agricultural Science
 - Combined Mathematics
 - Biology
 - Higher Mathematics
 - Chemistry
 - Physics
 - Information & Communication Technology

Details of the degree programmes under Applied Sciences (Physical Science) in different universities are given below:

Rajarata University of Sri Lanka:

- Bachelor of Science in Applied Sciences
Duration: 03 years

- Bachelor of Science Honours in Applied Sciences
Duration: 04 years
- Bachelor of Science Honours (Joint Major) in Applied Sciences
Duration: 04 years
- Bachelor of Science Honours in Industrial Mathematics
Duration: 04 years
- Bachelor of Science Honours in Chemistry
Duration: 04 years

Sabaragamuwa University of Sri Lanka:

- Bachelor of Science (Applied Sciences) Honours in Chemical Technology
- Bachelor of Science (Applied Sciences) Honours in Computer Science & Technology
- Bachelor of Science (Applied Sciences) Honours in Applied Physics
- Bachelor of Science (Applied Sciences) in Physical Sciences

University of Vavuniya, Sri Lanka:

- Bachelor of Science in Applied Mathematics & Computing
Duration: 03 years
- Bachelor of Science Honours in Computer Science
Duration: 04 years

Trincomalee Campus, Eastern University, Sri Lanka:

- Bachelor of Science in Applied Physics and Electronics
- Bachelor of Science Honours in Applied Physics
- Bachelor of Science Honours in Electronics

Wayamba University of Sri Lanka:

- Bachelor of Science (English Medium, 03 years)
- Bachelor of Science (Joint Major) in Major 1 and Major 2 † (English Medium, 04 years)
- Bachelor of Science Honours in Applied Electronics (English Medium, 04 years)
- Bachelor of Science Honours in Industrial Management (English Medium, 04 years)
- Bachelor of Science Honours in Mathematics with Statistics (English Medium, 04 years)

† Major 1 and Major 2 are two subjects from among the Computing & Information Systems, Electronics, Industrial Management, Mathematics & Mathematical Modeling and Statistics.

**2.2.4.8 Industrial Statistics & Mathematical Finance
(Course Code - 059)
(Proposed Intake - 121)**

Minimum eligibility requirements for admission :

At least 'S' grades in Combined Mathematics and any two subjects given below at the G.C.E. (Advanced Level) Examination;

- Higher Mathematics
- Physics
- Chemistry
- Information & Communication Technology

- Degree Programmes : Bachelor of Science
Duration : 03 years

Bachelor of Science Honours in Industrial Statistics
Duration : 04 years

Bachelor of Science Honours in Mathematical Finance
Duration : 04 years
- Available University : University of Colombo

**2.2.4.9 Statistics & Operations Research
(Course Code - 060)
(Proposed Intake - 79)**

Minimum eligibility requirements for admission :

At least 'S' grades in Combined Mathematics and two of the following subjects at the G.C.E. (Advanced Level) Examination;

- Biology
- Chemistry
- Physics
- Agricultural Science
- Mathematics
- Higher Mathematics
- Information & Communication Technology

- Degree Programme : Bachelor of Science Honours in Statistics and Operations Research
- Available University : University of Peradeniya
- Duration : 04 years

**2.2.4.10 Physical Science - Information and Communication Technology (Physical Science - ICT)
(Course Code - 108)
(Proposed Intake - 156)**

Minimum eligibility requirements for admission :

At least 'S' grades in Combined Mathematics, Physics and Information & Communication Technology at the G.C.E. (Advanced Level) Examination.

- Degree Programmes and Universities : Bachelor of Science in Physics and Electronics
University of Kelaniya

Bachelor of Science in Physics and ICT
[BSc (PICT)]
University of Sri Jayewardenepura
- Duration : 03 years

The respective Faculties of the University to which the student has been admitted will do the selection for a particular degree programme within a University.

**2.2.4.11 Artificial Intelligence
(Course Code - 117)
(Proposed Intake - 50)**

Minimum eligibility requirements for admission :

At least a 'C' grade in Combined Mathematics or Physics or Higher Mathematics;
and

At least 'S' grades in two other subjects listed below at the G.C.E. (Advanced Level) Examination;

- Combined Mathematics
 - Higher Mathematics
 - Mathematics
 - Physics
 - Chemistry
 - Information & Communication Technology
-
- Degree Programme : Bachelor of Science Honours in Artificial Intelligence [BScHons (AI)]
 - Available University : University of Moratuwa
 - Duration : 04 years
 - Medium : English

**2.2.4.12 Electronics and Computer Science
(Course Code - 119)
(Proposed Intake - 80)**

Minimum eligibility requirements for admission :

At least a 'C' grade in Combined Mathematics or Physics or Higher Mathematics
and;

Should have passed the three subjects at the G.C.E. (Advanced Level) Examination including Physics and Combined Mathematics or Higher Mathematics and the third subject from the list given below,

- Chemistry
 - Combined Mathematics
 - Higher Mathematics
 - Information & Communication Technology
-
- Degree Programmes : Bachelor of Science in Electronics and Computer Science [BSc (Elnc & Comp Sc)]
Duration : 03 years

Bachelor of Science Honours in Electronics and Computer Science [BScHons (Elnc & Comp Sc)]
Duration : 04 years

- Available University : University of Kelaniya
- Medium : English

2.2.4.13 Data Science
(Course Code - 136)
(Proposed Intake - 65)

Minimum eligibility requirements for admission :

At least a 'C' grade in Combined Mathematics or Physics or Higher Mathematics;
and

At least 'S' grades for two of the following subjects at the G.C.E. (Advanced Level) Examination;

- Combined Mathematics
- Physics
- Higher Mathematics
- Chemistry
- Information & Communication Technology

In addition to that, the candidates should have obtained at least a Credit Pass (C) in English at the G.C.E. (Ordinary Level) Examination.

- Degree Programme : Bachelor of Science Honours in Data Science [BScHons (Dat Sc)]
- Available University : Sabaragamuwa University of Sri Lanaka
- Duration : 04 years
- Medium : English

Candidates are required to submit the duly certified copy of the original G.C.E. (Ordinary Level) Examination certificate confirming the above grade along with the application for university admission. Failure to submit it will result in rejection of the application for the course of study in Data Science.

Candidates who have offered subjects in the Physical Science Stream at the G.C.E. (Advanced Level) Examination could also seek admission to the following courses of study offered by different universities, provided that they have fulfilled the entry requirements for those courses of study.

- Quantity Surveying
- Architecture
- Design
- Fashion Design & Product Development
- Law
- Management Studies (TV)
- Surveying Science
- Information Technology (IT)
- Management and Information Technology (MIT)

- Urban Informatics and Planning
- Peace and Conflict Resolution
- Information and Communication Technology (ICT)
- Facilities Management
- Science and Technology
- Computer Science & Technology
- Entrepreneurship and Management
- Physiotherapy
- Industrial Information Technology
- Mineral Resources and Technology
- Physical Education
- Sports Science & Management
- Management and Information Technology (SEUSL)
- Hospitality, Tourism and Events Management
- Information Technology & Management
- Tourism & Hospitality Management
- Information Systems
- Landscape Architecture
- Software Engineering
- Film & Television Studies
- Marine and Fresh Water Sciences
- Food Business Management
- Business Science
- Geographical Information Science
- Financial Mathematics and Industrial Statistics
- Human Resource Development
- Applied Chemistry
- Health Tourism and Hospitality Management
- Yoga and Parapsychology
- Social Studies in Indigenous Knowledge
- English Language and Applied Linguistics
- Creative Music Technology and Production
- Plantation Management and Technology
- Primary Education
- Polymer Science and Industrial Management